

MOOSE Project

- Three-part project
- Will upload input and output files to Moodle
- Will upload a written Part 1 report, max of 10 pages (including figures)
- This is an individual project, but some collaboration is encouraged
- Write up with deliverables from Part 1, choice of materials, mesh, details therein, etc.
- Part 1 is due March 3

MOOSE Project Part 1

- Fuel pellet dimensions listed
- This is a 1-D problem, but I want your geometry to be set up in 2-D RZ
- Assume reasonable values for material properties
- Outer cladding temperature is constant: 600 K
- Solve temperature profile for:
 - Steady-state: $Q = 350 \text{ W/cm}^2$
 - Compare against analytical solution
- Use both a constant k and a temperature-dependent k
- Perform a mesh convergence study
- Use transient solver
- Explore two mesh options:
 - explicitly meshing the gap
 - without meshing the gap

